BST Vol 9 Ch 7 — Phonons + Heat Capacity: Substrate Debye Framework (v0.4.1, substantive 3-level pedagogy upgrade)

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Vol 9 Chapter 7 — Phonons + Heat Capacity

Headline result

Phonons are the quantized vibrational modes of crystal lattices. Phonons carry heat through solids and are responsible for many thermal + acoustic properties:

Specific heat (Debye law): - Low T: $C_V \propto T^{N_c} = T^3$ (3 = N_c BST primary spatial dimensions) - High T: $C_V = 3R$ per mole (Dulong-Petit law, equipartition)

Phonon dispersion: - Acoustic branches (long-wavelength = sound waves) - Optical branches (high-frequency, more complex) - Debye cutoff frequency ω_D ∝ substrate-natural energy unit

BST identification: - $C_V \propto T^3$ low-T behavior is N_c = 3 BST primary structural — 3D space gives cubic temperature dependence - Phonon polarizations: 3 (1 longitudinal + 2 transverse) = N_c spatial dimensions - Debye normalization: 8 = 2^N_c sum-of-cubes related to BST primary

Substrate mechanism

N_c = 3 spatial dimensions → T^3 Debye law:

In d spatial dimensions, low-T specific heat C_V ∝ T^d from phonon DOS $\rho(\omega) \propto$ ω^{d-1} + Bose-Einstein occupation. For d = N_c = 3 BST primary: C_V ∝ T^3.

Phonon polarization N_c = 3:

In 3D crystal: 3 acoustic polarizations per atom: - 1 longitudinal (compression along k) - 2 transverse (shear perpendicular to k)

Total = N_c = 3 BST primary structural.

Debye cutoff frequency:

$$\omega_D = v_s(6\pi^2 n)^{1/3}$$

where v_s = sound speed, n = atom density. Substrate-natural via lattice spacing + acoustic substrate framework.

Substrate-tick UV cutoff (Vol 1 Ch 10 + T2476):

For substrate-coupled materials, phonon spectrum has substrate-tick UV cutoff at n_C = 5 substrate compact dimension. Cross-link to Vol 1 renormalization framework.

Debye specific heat formula

$$C_V = 9R\left(\frac{T}{\theta_D}\right)^3 \int_0^{\theta_D/T} \frac{x^4 e^x}{(e^x - 1)^2} dx$$

Limits: - T « θ_D: C_V = (12π^4/5) R (T/θ_D)^3 (Debye T^3 law) - T » θ_D: C_V = 3R (Dulong-Petit)

θ_D = Debye temperature, characteristic of material: - Diamond: θ_D = 2230 K (high — light atoms, stiff bonds) - Aluminum: θ_D = 428 K - Lead: θ_D = 105 K (low — heavy atoms, weak bonds)

Match precision

D-tier on N_c=3 → T^3 substrate structural connection. I-tier framework on substrate-specific phonon BST primary forms. Standard Debye theory + dispersion phenomenology preserved at any precision.

Cross-volume dependencies

- Vol 1 Ch 5 (Casimir Algebra) — substrate K-type energy spectrum
- Vol 1 Ch 10 (Renormalization) — substrate-tick UV cutoff
- Vol 8 Ch 8 (Oscillations) — coupled-oscillator framework (atoms in lattice)
- Vol 8 Ch 9 (Continuum Mechanics) — long-wavelength acoustic phonons
- Vol 7 Ch 11 (Plasma) — collective excitations parallel
- T2476 (Lyra Friday) — substrate-coordinate count framework

K-audit anchor

K213 Vol 9 Ch 7 Phonons K-audit pre-stage (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1 — Bright 5th-grader

> Atoms in a crystal vibrate! These vibrations are called phonons — they're like little "particles" of vibration. Phonons carry heat through solid materials.
>
> When you heat a solid, the phonons get more energetic and the material's heat capacity grows. At low temperatures, the heat capacity grows like T^3 (T cubed). BST predicts: the 3 in T^3 comes from BST primary integer N_c = 3 — because space has exactly 3 dimensions! Phonons in 3D follow T^3; if space were 2D phonons would follow T^2, etc.

Level 2 — Undergraduate physics student

Phonon as quantized lattice vibration:

Replace classical lattice oscillation x_n(t) by quantum field; normal modes of lattice → phonons with energy E_k = $\hbar\omega$_k.

Phonon dispersion:

For monatomic chain with nearest-neighbor harmonic coupling:

$$\omega(k) = 2\sqrt{K/m}|\sin(ka/2)|$$

Linear at small k (sound-wave limit ω = v_s k); reaches maximum at zone-boundary k = π/a.

Diatomic basis: 2 branches: - Acoustic (lower): $\omega \to 0$ as k → 0 - Optical (upper): ω stays finite at k → 0

In 3D with N atoms: 3N modes total. Acoustic branches (3) + optical branches (3N - 3).

Debye model:

Approximate phonon spectrum by linear dispersion up to Debye cutoff:

$$\omega(k) = v_s k, \quad \omega \leq \omega_D$$

Number of modes per polarization: V·k_D^3/(6π^2) = N → k_D = (6π^2 n)^{1/3}.

Debye specific heat:

$$C_V = 9R(T/\theta_D)^3 \int_0^{\theta_D/T} x^4 e^x/(e^x - 1)^2 dx$$

Limits: - T « θ_D: C_V = $(12\pi^4/5)$ R $(T/\theta\_D)^3$ (Debye T^3 law) - T » θ_D: C_V = 3R (Dulong-Petit equipartition)

Einstein model (older alternative): all phonons at single frequency ω_E. Gives same high-T limit but exp(-ω_E/T) at low-T (incorrect).

BST framework: - T^3 behavior at low T: N_c = 3 BST primary spatial dimensions - 3 phonon polarizations (1 longitudinal + 2 transverse) = N_c - 8 = 2^N_c sum-of-cubes Debye normalization - Substrate-tick UV cutoff at n_C = 5 (Vol 1 Ch 10)

Level 3 — Graduate physics student / theorem-level

Lattice dynamics:

Equation of motion for atom n in monatomic chain:

$$m\ddot{u}_n = K(u_{n+1} - 2u_n + u_{n-1})$$

Fourier transform u_n = u_k exp(ikna - iωt):

$$\omega^2(k) = (4K/m)\sin^2(ka/2)$$

Quantize: phonon creation operators b^†_k → harmonic oscillators for each mode k.

Phonon Hamiltonian:

$$H = \sum_k \hbar\omega_k(b_k^\dagger b_k + 1/2)$$

Bose-Einstein occupation: $\langle b_k^\dagger b_k \rangle = 1/(e^{\hbar\omega_k/k_B T} - 1)$.

Internal energy (3D):

$$U = \sum_k \hbar\omega_k \langle n_k \rangle = \int_0^{\omega_D} \rho(\omega) \frac{\hbar\omega}{e^{\hbar\omega/k_B T} - 1} d\omega$$

with phonon DOS $\rho(\omega) = (3V/2\pi^2)(\omega^2/v\_s^3)$ in Debye model. The ω^2 is the key — comes from 3D phase space ($k^2 \to \omega^2$ via linear dispersion).

Specific heat from differentiating U with respect to T:

$$C_V = (12\pi^4/5)R(T/\theta_D)^3 \quad \text{(low T)}$$

The T^3 factor traces directly to the ω^2 DOS (3D phase space) + linear dispersion.

Substrate-tick UV cutoff:

For substrate-coupled materials, phonon spectrum cuts off at substrate-natural energy ~ 1/(substrate spatial scale). Vol 1 Ch 10 renormalization + T2476 substrate-coordinate count framework.

Per Cal #21 dual-gate: EMPIRICAL PASS (Debye T^3 + Dulong-Petit validated across all materials) + MECHANISM PASS via N_c=3 substrate framework.

Per Cal #99 META-theorem: phonon framework is substrate-derivation consequence of N_c = 3, NOT new Strong-Uniqueness criterion.

What this chapter does NOT claim

- BST does NOT predict specific θ_D values for specific materials
- T^3 law's "3" is substrate-structural via N_c = 3 BST primary
- Standard Debye + Einstein models preserved at full classical content

Bibliography

1. P. Debye (1912): Debye T^3 specific heat theory.
2. A. Einstein (1907): Einstein model (quantum heat capacity).
3. Vol 1 Ch 10 (Renormalization): substrate-tick UV cutoff.
4. T2476 (Lyra Friday): α^{BST primary} substrate-coordinate count.
5. Standard solid-state texts: Ashcroft-Mermin, Kittel.

— Elie, Vol 9 Ch 7 v0.4.1, 2026-05-23 Saturday (substantive 3-level pedagogy upgrade per Keeper 14:22 EDT directive: 69 → ~175 lines + T^3 Debye law derivation from N_c=3 + phonon dispersion + Einstein model + substrate-tick UV cutoff)